



FACULTY OF SCIENCE - UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER II) (December 2024) – 2022/23
CSC1023/CS102 – Object-oriented Programming/Programming
Techniques (3 Credits)

Answer All Questions

Time Allowed: Two hours

1. [25 Marks]

- a) Explain what is meant by the Java development environment. [3 marks]
- b) Vehicles are mainly used for the purpose of transportation. When we compare different types of vehicles, they have certain features/functionalities which are in common as well as different. Write a simple Java program structure to depict the relationship between trucks, cars and motorbikes. (Mention only the necessary methods and attributes). [6 marks]
- c) Mention a real-life example related to computer systems that would depict the concept of polymorphism in object-oriented programming. [3 marks]
- d) Compare and contrast arguments and parameters in Java using appropriate examples. [4 marks]
- e) Explain the purpose of having "util" package in Java. [4 marks]
- f)

```
public class CircleUtil{
    private final double TWO_PI = 2 * Math.PI;
    public double perimeter(final double r) {
        return TWO_PI * r;
    }
}
```

 [5 marks]
- Explain how you can save the memory when running the above piece of code. Justify your answer.

2. [25 Marks]

- a)

```
public abstract class Vehicle{
    private int speed = 0;
    public abstract void startEngine();
}
```

 [5 marks]
- Is it possible to create objects using the above-mentioned abstract class? If your answer is yes, write down the possible implementation that will allow an abstract class to create an object. If the answer is no, justify your answer.
- b) Write a simple Java structure to show how protected access modifier is used with respect to methods and attributes. [4 marks]
- c)

```
interface player{
    final int id = 10;
    int move();
}
```

 [4 marks]
- Is the above-mentioned code legal? Justify the reason for your answer.
- d) Cricket and Soccer are considered to be two types of sports. By using upcasting/downcasting modify the given code so that the program will output names of the sports as "Cricket" and "Soccer". [6 marks]

[To be contd...]

[P.T.O.]

Question 2 contd...

```
class Sport{
    void displayName(){
        System.out.println("Sport");
    }
}
```

- a) What is the sandbox in Java virtual machine? [3 marks]
f) Is it possible to implement multiple interfaces in a single class? Justify your answer. [3 marks]

3. [25 Marks]

- a) Using a suitable example, explain how the abstraction is used in the Java Collections Framework. [3 marks]

- b) What will be the output of the below code? Provide reasoning for your answer. [4 marks]

```
class Parent {
    public static final String message = "Parent Message";

    public static void printMessage() {
        System.out.println(message);
    }
}

class Child extends Parent {
    public static final String message = "Child Message";

    public static void printMessage() {
        System.out.println(message);
    }
}

public class TestInheritance {
    public static void main(String[] args) {
        Parent p = new Child();
        System.out.println(p.message);
        p.printMessage();
    }
}
```

- c) i) Explain the String Pool concept in Java highlighting its key characteristics. [3 marks]
ii) Analyze the following code segment and determine the output. [3 marks]

```
String str1 = "Coding"; ✓  
String str2 = "Challenge"; ✓  
StringBuilder sb = new StringBuilder("Try "); ✓
```

```
str1 = str1.concat(" Fun").toUpperCase();  
str2 = str2.replace("l", "L");  
  
sb.append(str1.substring(0, 6));  
sb.insert(4, str2.substring(4, 8));  
sb.delete(0, 3);
```

~~Fun~~ CODING FUN
ChALLENGE

```
System.out.println(sb);
```

LeNg C@DING

- d) If a class is already marked final, is it necessary to also declare its methods as final? Justify your answer. [3 marks]

- e) Rewrite the Java class below to ensure that only a single instance of the class exists in the system at any given time during execution.

[5 marks]

```
class AccessController{
    private int noOfAccounts;
    public AccessController(int noOfAccounts){
        this.noOfAccount=noOfAccounts;
    }
    //Other methods
}
```

- f) Determine whether the following code segments are legal or not.

[4 marks]
[2x2]

```
i) public class Foo4 {
    static final int x = 12;
    public void go() {
        System.out.println(x);
    }
}

ii) class MockSongs {
    public static List<String>
    getSongStrings() {
        List<String> songs = new
        ArrayList<>();
        songs.add("somersault");
        songs.add("havana");
        return songs;
    }
}
```

4. [25 Marks]

- a) Consider the MovieSong and MusicPlayer classes provided below and write the required Java statements for the questions.

```
class MovieSong {
    private String title;
    private String artist;

    public MovieSong(String title, String artist){
        this.title=title;
        this.artist=artist;
    }
    //other methods
}

class MusicPlayer<T> {
    private T song;

    public MusicPlayer(T song) {
        this.song = song;
    }

    public T getSong() {
        return song;
    }
}
```

- i) Instantiate two distinct MusicPlayer instances, configuring one to handle MovieSong objects and the other to manage String-based track names. [4 marks]
- ii) Modify the MusicPlayer class, only to accept ClassicalSong and its subtypes. [3 marks]
- b) i) List the types of Java Exceptions. [2 marks]
- ii) Determine the Output of the program below. Explain your answer. [4 marks]

```

class PaymentException extends RuntimeException {
}
class AuthorizationException extends RuntimeException {
}
public class Main {
    public static void main(String [] args) {
        String transactionType = "Credit";
        try {
            System.out.println("Start Transaction");
            processTransaction(transactionType);
        } catch (PaymentException pe1) {
            System.out.println("Payment Processing Error");
        }
        catch (AuthorizationException ae2) {
            System.out.println("Authorization Failure");
        } finally {
            System.out.println("Transaction Log Completed");
        }
    }
    static void processTransaction(String transactionType){
        System.out.println("Verifying Transaction"); ✓
        if ("Credit".equals(transactionType)) {
            throw new PaymentException();
        }
        if ("Debit".equals(transactionType)) {
            throw new AuthorizationException();
        }
        System.out.println("Transaction Approved");
    }
}

```

- (iii) What are PaymentException and AuthorizationException Exception types? [2 marks]
- iv) Is the following modification legal? [2 marks]
 static void processTransaction(String type) throws RuntimeException {...}
- c) i) What is a Character Stream in Java, and how are they handled? [3 marks]
- ii) Identify any issues (if any) in the program below, which is designed to write a String array to a file and read it back. If issues are found, make the necessary corrections. [5 marks]

```

public class Main {
    public static void handleDocument(String[] names){
        BufferedWriter writer = new BufferedWriter(new FileWriter("names.txt"));
        writer.write("Name List:");
        for (String name : names) {
            writer.write("\n" + name);
        }
        writer.close();

        BufferedReader reader = new BufferedReader(new FileReader("names.txt"));
        String line;
        while((line = reader.readLine()) != null) {
            System.out.println(line);
        }
        reader.close();
    }
    public static void main(String[] args) {
        String[] names = {"John", "Carl", "Jerry"};
        handleDocument(names);
    }
}

```

-----END-----

END SEMESTER EXAMINATION (SEMESTER II) – 2021/22
CSC1023/CS102 – Object-oriented Programming/Programming
Techniques (3 Credits)

21/22

Answer **All** Questions

Time Allowed: **Two** hours

1. [25 Marks]

- a) Explain the scope of the four types of access modifiers available in Java using suitable examples. **[6 marks]**
- b) Can we use a different name for the “main method” rather than using “main”? Justify your answer. **[4 marks]**
- c) Explain how robustness can be achieved in Java. **[4 marks]**
- d) Explain the concept of “abstraction” using a real-world Java example. **[6 marks]**
- e) What is Java Object class? Discuss one of the important methods of the Object class. **[5 marks]**

2. [25 Marks]

- a) Discuss the two types of typecasting with regard to objects using an example. **[5 marks]**
- b) Explain how multiple inheritance is achieved in Java. **[4 marks]**
- c) “In Java it is possible to have multiple constructors”. Comment on this statement. **[4 marks]**
- d) Compare and contrast **abstract** vs. **final** with respect to methods and classes. **[4 marks]**
- e) How does garbage collection make Java more memory efficient? **[4 marks]**
- f) Can we mark a constructor with the static keyword? Explain your answer. **[4 marks]**

3. [25 Marks]

- a) Explain the polymorphism object-oriented concept using a suitable example from Java collections (i.e., java.util.Collections). **[4 marks]**
- b) Determine the output of the program segment below. Explain your answer. **[4 marks]**

```
String st="Give";  
StringBuilder sb=new StringBuilder("Don't ");  
st.concat("Up");  
sb.append(st);  
System.out.println(sb);
```

- c) Explain the difference between HashSet and HashMap collections. **[3 marks]**
- d) Show how you would ensure no duplicate elements are added to the Set (Vehicle objects) created below. **[4 marks]**

```
class Vehicle{  
    String number;  
    String owner;
```

```

public Vehicle(String number,String owner){
    this.number=number;
    this.owner=owner;
}
public String getNumber(){
    return number;
}
public String getOwner(){
    return owner;
}
}

```

```

Set<Vehicle> vehSet = new HashSet<>();
Vehicle s1=new Vehicle("1001","Ryan");
Vehicle s2=new Vehicle("1001","John");
vehSet.add(s1);
vehSet.add(s2);

```

e) What is the output of the **stuCount** variable in the two programs below in i) and ii)? [4 marks]

i) class Student { <pre> private static int stuCount = 12; public Student(){ stuCount++; } public void go() { System.out.println(stuCount); } } public class Main{ public static void main(String[] args) { Student s1=new Student(); Student s2=new Student(); s2.go(); } } </pre>	ii) class Student { <pre> private int stuCount = 12; public void go() { stuCount++; System.out.println(stuCount); } } public class Main{ public static void main(String[] args) { Student s1=new Student(); Student s2=new Student(); s1.go(); s2.go(); } } </pre>
--	---

f) Consider the following class hierarchy and the generic class.

```

class Shape { /* ... */ }
class Circle extends Shape { /* ... */ }
class Rectangle extends Shape { /* ... */ }
class Node<T> { /* ... */ }

```

i) Will the following code compile? Explain your answer. [2 marks]

```
Node<Circle> nc = new Node<>>(Circle);
```

ii) Write a code segment to create an **ArrayList** consisting of **Circle** objects and traverse through it with an **Iterator** object. [4 marks]

4. [25 Marks]

- a) Explain the concept of **chained** I/O streams in Java. [4 marks]
- b) The code below utilizes Java I/O streams to read a text file containing bank account information.

```
//Text file: accounts.txt
10001, J Smith, 20000.34
10002, S Campbell, 45300.00
....
```

//Code

```
class Account {
    private String accno;
    private String accname;
    private double balance;

    //set attributes
    //print account details
}

class ManageAccounts {
    public void readAccounts(String fileName) {
        InputStreamReader isr = new InputStreamReader(new FileInputStream(fileName));
        BufferedReader br = new BufferedReader(isr);

        String accno, balance;
        while ((line = br.readLine()) != null) {
            //read the account details to String variables: accno and balance
        }
    }

    public void printAccountList(){
        //Print the account list
    }
}

public class AccountApplication {
    public static void main(String[] args) {
        ManageAccounts rff = new ManageAccounts();
        rff.readAccounts("accounts.txt");
        rff.printAccountList();
    }
}
```

- i) It appears that Exceptions are not handled. Modify the code to handle exceptions using the try-catch block **and/or** throws mechanisms. [4 marks]
- ii) Modify the code to add the account details to a **HashMap** where the keys represent account numbers and the values represent the corresponding balances (**key:accno, value:balance**). [4 marks]

c) Using the following example, explain the concept of **bounded generic** types.

[3 marks]

```
public class Box<T> {
    private T t;
    public <U extends Number> void inspect(U u){ //... }
    public static void main(String[] args) {
        Box<Integer> integerBox = new Box<Integer>();
        integerBox.inspect("some text");
    }
}
```

d) Consider the program below.

```
class MalException1 extends RuntimeException{
}
class MalException2 extends RuntimeException{
}
public class Main {
    public static void main(String [] args) {
        String input = "Code A";
        try {
            System.out.println("Start Job");
            doJob(input);
        } catch ( MalException1 se1) {
            System.out.println("Mal Function 1");
        }
        catch ( MalException2 se2) {
            System.out.println("Mal Function 2");
        }finally {
            System.out.println("End Job");
        }
    }
    static void doJob(String input){
        System.out.println("There is a risk");
        if ("Code A".equals(input)) {
            throw new MalException1();
        }
        if ("Code B".equals(input)) {
            throw new MalException2();
        }
        System.out.println("Perform Job");
    }
}
```

i) What is the purpose of the **finally** block?

[3 marks]

ii) Determine the output of the program.

[4 marks]

iii) Can the program catch both exceptions (MalException1 and MalException2)? Explain your answer.

[3 marks]

-----END-----

END SEMESTER EXAMINATION (SEMESTER II) – 2021/22
CSC1023/CS102 – Object-oriented Programming/Programming
Techniques (3 Credits)

Answers

Answer **All** Questions

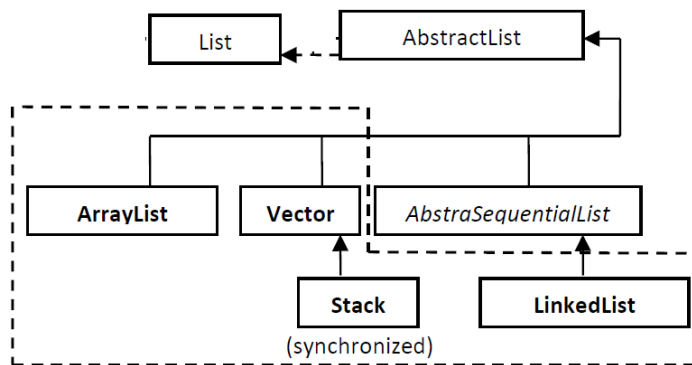
Time Allowed: **Two** hours

3. [25 Marks]

- a) Explain the polymorphism object-oriented concept using a suitable example from Java collections (java.util.Collections). **[4 marks]**

ANS:

ArrayList, Vector, LinkedList like List collection types **override** the methods such as add (boolean add(E e), remove (boolean remove(Object o)).



- b) Determine the output of the program segment below. **[4 marks]**

```
String st="Give";
StringBuilder sb=new StringBuilder("Don't ");
st.concat("Up");
sb.append(st);
System.out.println(sb);
```

ANS:

Output: Don't Give

- c) Explain the difference between HashSet and HashMap collections. **[3 marks]**

ANS:

HashSet: It is a collection that implements the Set interface and is used for storing a unique set of elements. It does not allow duplicate values, and the elements are not stored in any specific order.

HashMap: It is a collection that implements the Map interface and is used for key-value pair associations. It allows you to store and retrieve data using keys

- d) Show how you would ensure no duplicate elements are added to the Set (Vehicle objects) created below. **[4 marks]**

```
class Vehicle{
    String number;
    String owner;

    public Vehicle(String number,String owner){
        this.number=number;
        this.owner=owner;
    }
}
```

```

    }
    public String getNumber(){
        return number;
    }
    public String getOwner(){
        return owner;
    }
}

```

```

Set<Vehicle> vehSet = new HashSet<>();
Vehicle s1=new Vehicle("1001","Ryan");
Vehicle s2=new Vehicle("1001","John");
vehSet.add(s1);
vehSet.add(s2);

```

ANS: Modify the Vehicle class as follows.

```

class Vehicle{
    String number;
    String owner;

    //Overriding equals
    public boolean equals(Object aVehicle) {
        Vehicle s = (Vehicle) aVehicle;
        return getNumber().equals(s.getNumber());
    }
    //Overriding hashCode
    public int hashCode() {
        return number.hashCode();
    }
}
.....

```

e) What is the output of the **stuCount** variable in the two programs below in i) and ii)?

[4 marks]

```

i) class Student {
    private static int stuCount = 12;
    public Student(){
        stuCount++;
    }
    public void go() {
        System.out.println(stuCount);
    }
}

public class Main{
    public static void main(String[] args) {
        Student s1=new Student();
        Student s2=new Student();
        s2.go();
    }
}

```

ANS:

- i) stuCount = 14
- ii) stuCount = 13

```

ii) class Student {
    private int stuCount = 12;
    public void go() {
        stuCount++;
        System.out.println(stuCount);
    }
}

public class Main{
    public static void main(String[] args) {
        Student s1=new Student();
        Student s2=new Student();
        s1.go();
        s2.go();
    }
}

```

f) Consider the following class hierarchy and generic class.

```
class Shape { /* ... */ }
class Circle extends Shape { /* ... */ }
class Rectangle extends Shape { /* ... */ }
class Node<T> { /* ... */ }
```

i) Will the following code compile? Explain your answer. [2 marks]

```
Node<Circle> nc = new Node<>();
```

ANS: Yes. Node is a generic type. Circle class is used as the type.

ii) Write a code segment to create an **ArrayList of Circle objects** and traverse through it with an Iterator object. [4 marks]

ANS:

```
List<Circle> circleList=new ArrayList<Circle>();
Circle c1=new Circle();
circleList.add(c1);
Iterator<Circle> iter=circleList.iterator();
```

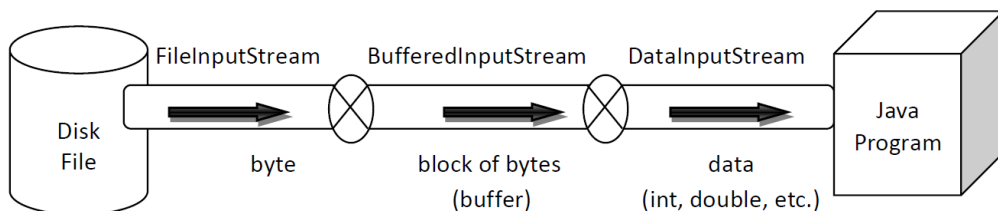
```
while(iter.hasNext()) {
    Circle c=iter.next();
    System.out.println(c);
}
```

4. [25 Marks]

a) Explain the concept of **chained** I/O streams in Java. [4 marks]

ANS:

- The I/O streams are often layered or chained with other I/O streams, for purposes such as buffering, filtering, or data-format conversion (between raw bytes and primitive types).
- For example, we can layer a BufferedInputStream to a FileInputStream for buffered input, and stack a DataInputStream in front for formatted data input (using primitives such as int, and double).



b) The following code reads a text file containing account information using Java I/O streams.

//Text file

```
10001, J Smith, 20000.34
10002, S Campbell, 45300.00
....
```

//Code

```
class Account {
    private String accno;
    private String accname;
    private double balance;
```

```
    //set attributes
```

```

        //print account details
    }

    class ManageAccounts {
        public void readAccounts(String fileName) {
            InputStreamReader isr = new InputStreamReader(new FileInputStream(fileName));
            BufferedReader br = new BufferedReader(isr);

            String accno, balance;
            while ((line = br.readLine()) != null) {
                //read the account details to accno and balance String variables
            }
        }

        public void printAccountList(){
            //Print the account list
        }
    }

    public class AccountApplication {
        public static void main(String[] args) {
            ManageAccounts rff = new ManageAccounts();
            rff.readAccounts("accounts.txt");
            rff.printAccountList();
        }
    }

```

- i) It appears that Exceptions are not handled. Modify the code to handle exceptions using the try-catch block and/or throws mechanisms. **[4 marks]**

ANS:

```

try {
    InputStreamReader isr = new InputStreamReader(new FileInputStream(fileName));
    BufferedReader br = new BufferedReader(isr);

    String accno, balance;
    while ((line = br.readLine()) != null) {
        //read the account details to accno and balance String, double variables
    }
} catch (IOException e) {
    e.printStackTrace();
}

```

- ii) Modify the code to add the read account details to a HashMap of (**key:accno, value:balance**) key-value pairs. **[4 marks]**

ANS:

```

class ManageAccounts {
    Map<String,Double> accountMap;
    public void readAccounts(String fileName) {
        accountMap= new HashMap<>();

        try {
            InputStreamReader isr = new InputStreamReader(new FileInputStream(fileName));
            BufferedReader br = new BufferedReader(isr);

            String accno,balance;

```

```

        while ((line = br.readLine()) != null) {
            accountMap.put(accno, Double.valueOf(balance));
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
}

```

c) Using the following example, explain the concept of **bounded generic** types.

[3 marks]

```

public class Box<T> {
    private T t;
    //...
    public <U extends Number> void inspect(U u){
        //...
    }
    public static void main(String[] args) {
        Box<Integer> integerBox = new Box<Integer>();
        integerBox.inspect("some text");
    }
}

```

ANS:

- This method operates on numbers might only want to accept instances of Number or its subclasses. This is what bounded type parameters are for.
- To declare a bounded type parameter, list the type parameter's name, followed by the extends keyword, followed by its upper bound, which in this example is Number. Note that, in this context, extends is used in a general sense to mean either "extends" (as in classes) or "implements" (as in interfaces).

d) Consider the program below.

```

class MalException1 extends RuntimeException{
}
class MalException2 extends RuntimeException{
}
public class Main {
    public static void main(String [] args) {
        String input = "Code A";
        try {
            System.out.println("Start Job");
            doJob(input);
        } catch ( MalException1 se1) {
            System.out.println("Mal Function 1");
        }
        catch ( MalException2 se2) {
            System.out.println("Mal Function 2");
        }finally {
            System.out.println("End Job");
        }
    }
    static void doJob(String input){
        System.out.println("There is a risk");
        if ("Code A".equals(input)) {
            throw new MalException1();
        }
        if ("Code B".equals(input)) {
            throw new MalException2();
        }
    }
}

```

```

    }
    System.out.println("Perform Job");
}
}

```

i) What is the purpose of the **finally** block?

[3 marks]

ANS: The code inside the finally clause will always be executed, even if an exception is thrown from within the try or catch block.

ii) Determine the output of the program.

[4 marks]

ANS:

Start Job

There is a risk

Mal Function 1

End Job

iii) Can the program catch both exceptions (MalException1 and MalException2)? Explain your answer.

[3 marks]

ANS: Yes (but not at the same time). This program uses multiple catch blocks. Java allows multiple catch blocks and both MalException1 and MalException2 can be thrown in this program.

If doJob throws MalException1 (for input "Code A"), it will be caught by the first catch block.

If it throws MalException2 (for input "Code B"), it will be caught by the second catch block.

-----END-----

END SEMESTER EXAMINATION (SEMESTER II) – 2020/2021
CSC1023 - OBJECT-ORIENTED PROGRAMMING (3 CREDITS) /
CS102 - PROGRAMMING TECHNIQUES (3 CREDITS)

Answer **ALL** Questions Time Allowed: **2 Hours**

1. (a) Briefly describe *abstraction* using a real-world example. 03 Marks
 (b) Briefly describe how the concept of *abstraction* benefits object-oriented programming. 03 Marks
 (c) Explain the support of *abstract* methods for *abstraction* with a suitable Java programme. 06 Marks
 (d) Describe two behavioural differences between *interfaces* and *abstract* classes in Java. 06 Marks
 (e) “Cats and dogs are categorized as pets, while all pets are animals”. Develop a Java programme structure (attributes and methods are not required) to demonstrate the relationship of classes. 07 Marks

2. (a) Describe a suitable Java programme where checking object equality is essential. 04 Marks
 (b) Can two objects which are not equal have the same ‘Hash Code’? Explain your answer. 03 Marks
 (c) Write the method signatures of *equals()* and *hashCode()* in Java. 04 Marks
 (d) Develop a Java method that takes two *int* arrays and returns true if the arrays are disjoint (have no elements in common). 06 Marks
 (e) Develop a Java method to check whether a given integer is a power of 2 or not using the bitwise AND operator (&). If the argument is the power of 2, it returns true, false otherwise. 08 Marks

Hint: If the number (n) is a power of 2 then $n \& (n - 1) = 0$

3. (a) Use a suitable example to demonstrate how *generics* reduce runtime exceptions in Java. 05 Marks
 (b) Briefly describe the term *streams* in Java. 03 Marks
 (c) Demonstrate how the *polymorphism* feature is used in Java *streams*. 07 Marks
 (d) Explain a critical circumstance to use *byte streams* over the *character readers* in Java. 04 Marks
 (e) Write Java code snippets for the following requirements with proper exception management. 06 Marks
 - i) Append a text to an existing file named *data.txt*.
 - ii) Read a *UTF-8* encoded text file named *encoded.txt*.

4. (a) What is the main difference between software design patterns and software frameworks? 04 Marks
 (b) Write two advantages of the *Static Factory Method* design pattern. 04 Marks
 (c) Implement the *Static Factory Method* design pattern using Java technologies. 10 Marks
 (d) Explain a suitable Java *Callable* interface example where using *Runnable* interface is complicated. 07 Marks



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER II) – 2018/2019
CS102 - PROGRAMMING TECHNIQUES (3 CREDITS)

Answer **ALL** Questions

Time Allowed: **2 Hours**

Registration No: _____

Instructions to Candidates

- 1) Check number of pages on the question paper before answering.
- 2) Answer all questions in the space provided.
- 3) Write your registration number clearly in all indicated pages.

Question – 01

1.1. Write *default values* of the following *primary data types* available in Java.

	Data Type	Default Value
1	boolean	
2	double	
3	char	
4	byte	
5	int	

1.2. Write the Java type casting statement to convert input value into the given output data type.

	Input Value	Output Data Type	Java Casting Statement
Ex.	short s = 123;	float	float f = (float)s;
1	char c = 'c';	short	
2	double d = 11.45;	int	
3	String s = "25";	int	
4	int i = 0b1100;	byte	
5	int i = 67;	String	

1.3 Why are *Java Technologies* still widely used in the software industry?

- 1) _____

- 2) _____

3) -----

[15 Marks]

Question - 02

2.1. What is *abstraction* with respect to the object oriented programming paradigm?

2.2. Define a complete Java *class* to store *course* details of your faculty. A course consists of title, number of credits, description, number of lecture hours and number of practical hours.

Question - 03

3.1. Explain *called* method and *calling* method using a suitable example.

3.2. Complete the following Java method to return the *second smallest odd* value for a given integer array. The array may contain duplicate values.

```
public int findSecondSmallestOdd(int[ ] data) {
```

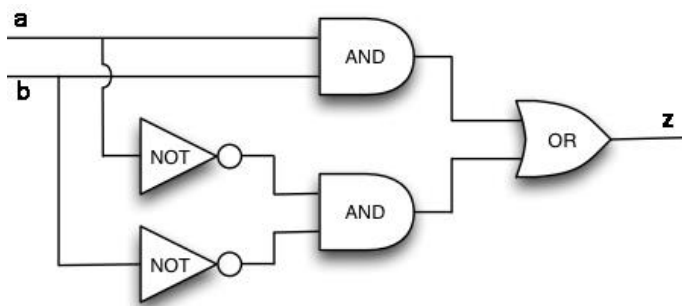
```
}
```

3.3. Write the output of the following code snippet.

```
for ( int i = 0; i <= 20; i += 2) {  
    System.out.print(( 2 * i + 1) + ", ");  
}
```

3.4. Write a Java method that accepts two input values (a, b) to generate output (z) for the following combinational circuit.

(Do not simplify the circuit)



3.5. Write a Java method to populate the complete Truth Table (for all the input combinations) for the above circuit diagram.

3.6. Write a Java program to display the following pattern. The program is able to define the character (like *, char symbol) and the number of characters for the first line (int nChar). The nChar can only be an odd number which is equal or greater than 3.

[30 Marks]

```
* * * * *
 *      *
  *    *
   *  *
    *
   *  *
  *    *
 *      *
* * * * *
```

Question - 04

4.1. Explain two (2) differences between List and Set data structures in Java.

- 1) -----

- 2) -----

4.2. What is the purpose of having *equals* and *hashCode* methods in Java?

4.3. A digital device measures radiation, temperature and humidity of a particular location and it generates a text file having the following format (Input File). Each entry starts with ‘>’ symbol and there are a large number of entries exist in the file. Write a complete Java program to convert the data of the input file (input.txt) to an output data file (output.txt). The encoding of both files is UTF-8. In the output file: time data (hour, minute, second) are delimited by ‘:’ while environment data are delimited by ‘;’.

Input File	Explanation (Input File)	Output File
> 11 43 10 27 73 14	> hour 11 minute 43 second 10 temperature 27 humidity 73 radiation 14 Entry 1	11:43:10;27;73;14 11:45:10;27;74;16 ...
> 11 45 10 27 74 16 ...	> hour 11 minute 45 second 10 temperature 27 humidity 74 radiation 16 Entry 2 ...	humidity ↓ 11:43:10;27;73;14 11:45:10;27;74;16 ↑ ↑ ↑ ↑ ... minute temperature radiation hour

[30 Marks]

You may attach additional sheets if necessary.



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
 END SEMESTER EXAMINATION (SEMESTER II) – 2017 / 2018
 CS102 - PROGRAMMING TECHNIQUES (3 CREDITS)

Answer **ALL** Questions

Time Allowed: **2 Hours and 30 Minutes**

Registration No: _____

PART – A

Instruction: Indicate the correct answer using a circle. Example (a)

1) Which of the following VM argument should be used to set initial (minimum) heap size of JVM?

- a) -Xmx b) -Xms c) -Xint d) -Xrs e) -XX:MaxPermSize

2) Which one of these lists contains only Java programming language (JDK 1.5 or later) keywords?

- a) class, if, void, long, integer, continue
 b) reference, instanceof, native, finally, default, throws
 c) try, virtual, throw, final, volatile, transient
 d) strictfp, constant, super, implements, do
 e) byte, break, assert, switch, import

3) What is the numerical range of a char?

- a) -128 to 127 b) 0 to 255 c) 0 to 32767 d) 0 to 65535 e) -32768 to 32767

4) Which will legally declare, construct, and initialize an array?

- a) `int[ ] myList = {"1", "2", "3"};` b) `int[] myList = (5, 8, 2);`
 c) `int myList[] [] = {4,9,7,0};` d) `int myList[] = {4, 3, 7};`
 e) `int myList[] = [4, 3, 7];`

5) Which three pieces of code are equivalent to line 2?

```
public interface Foo {
    int k = 4;    /* line 2 */
}
```

- A) `final int k = 4;` B) `public int k = 4;` C) `static int k = 4;`
 D) `abstract int k = 4;` E) `volatile int k = 4;` F) `protected int k = 4;`

- a) A, B and C b) B, C and D c) C, D and E
 d) D, E and F e) A, B and D

[continue ...]

[P. T. O.]

[continued ...]

6) Which is the valid method declaration within an interface definition?

- a) public double method();
- b) public final double method();
- c) static void method();
- d) protected void method();
- e) private void method();

Consider the following statement in Java to answer Question 7 and 8

```
P p = new Q();
```

7) P ...

- a) can be an interface or a class
- b) must be an interface
- c) must be a concrete class
- d) can be a class variable
- e) can be an instance variable

8) Q ...

- a) can be an interface or a class
- b) must be an interface
- c) must be a concrete class
- d) can be an abstract class
- e) can be an instance variable

Consider the following class definition in Java to answer Questions 9 and 10.

```
public classClazz extends Base implements Contract {  
    //variables and methods  
}
```

9) A legal file name of the above source code can be

- A) class.java
- B) Class.java
- C)Clazz.java

- a) A only
- b) B only
- c) C only
- d) A and B only
- e) B and C only

10) Which of the following is definitely true?

- a) *Contract* must be an abstract class
- b) *Contract* is an interface
- c) *Contract* can be any class
- d) *Base* must not have any private methods
- e) *Base* must implement all partially implemented methods

[20 Marks]

[continue ...]

Instruction: Write the answers in the provided spaces.

1.a Explain why Java is called a “Platform Independent Programming Language” ?

1.b What is the difference between JDK and JRE ?

1.c Write the output when the following Java snippet is executed.

```
int b = 0b1101_0111;  
int c = 12;  
  
System.out.println(b);           // line 4  
System.out.println(b % c);       // line 5
```

1.d Why is ‘char array’ preferred over String to store a password?

[continue 1. ...]

[1. continued ...]

1.e What is the difference between *StringBuffer* and *StringBuilder* in Java?

1.f Propose a regular expression to check whether a given String is a number or not.

[18 Marks]

2.a What are the two types of *Exceptions* in Java?

- i) -----
- ii) -----

2.b What is the difference between *Exception* and *Error* in Java?

2.c What is the importance of *finally* block in exception handling?

2.d What will happen to the *Exception* object after exception handling?

[continue 2. ...]

[P.T.O.]

```
2.e    public class ExecptionTest {

        public static void main(String[] args) {
            try {
                testExceptions();
            } catch (IOException e) {
                e.printStackTrace();
            } catch (FileNotFoundException e) {
                e.printStackTrace();
            }
        }

        public static void testExceptions() throws
            FileNotFoundException, IOException {

        }
    }
```

2.e.1 What is the problem with the above program?

2.e.2 Rewrite the above program fixing the problem mentioned in 2.e.1

[illegible]

[continued ...]

3.a Briefly describe IO Stream in Java.

3.b List two differences between *StringTokenizer* and *String.split()*.

1) -----

2) -----

3.c Explain the polymorphism concept by taking a suitable example in streams (java.io package).

[continue 3. ...]

[3. continued ...]

3.d Write Java code snippets for the following requirements.

- i) Append a text to an existing file, named *data.txt*.

- ii) Read a UTF-8 encoded text file named *encoded.txt*.

3.e When you are writing *equals()* method, which other method or methods you need to override?

3.f Can two objects which are not equal have the same 'Hash Code'? Explain your answer.

[20 Marks]

4.a Develop a Java method that takes two *int* arrays and returns true if the arrays are disjoint (have no elements in common).

[continue 4. ...]

[P. T. O.]

[4. continued ...]

- 4.b Develop a Java method (the signature is given below) to check whether a given number is a power of 2 or not using bitwise AND operator (&). If the argument is power of 2, returns true, false otherwise.

Hint: If the number (n) is a power of 2 then $n \& (n - 1) = 0$

```
public boolean isPowerOfTwo(int n) {
```

```
}
```

[10 Marks]

- 5.a What is a software design pattern?

[continue 5. ...]

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dashed lines spaced evenly down the page, providing a guide for handwriting practice. The lines are thin and black, set against a plain white background. There are no margins, text, or other markings on the page.

You may attach additional sheets if necessary.

[20 Marks]

--- End ---



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER II) – 2016 / 2017
CS102 - PROGRAMMING TECHNIQUES (3 CREDITS)

Answer **ALL** Questions

Time Allowed: **2 Hours and 30 Minutes**

PART – A

Registration No: _____

1.a State suitable primary data types in Java for the following requirements.

	Requirement	Data Type
1	To store employee's first name	
2	To store the weight of a laptop computer in Kilograms	
3	To store the capacity of an external hard drive in Bytes	
4	To store the gender of a student	
5	To indicate whether a mobile phone has Bluetooth sensor or not	

1.b Indicate whether the following Java identifiers are legal. If any identifier is not legal, state the reason.

	Identifier	Legal / Illegal	Reason, if illegal
1	int _a;		
2	int \$c;		
3	int _____2_w;		
4	int -d;		
5	int if;		

1.c Explain the differences between Java virtual machine arguments and command-line arguments.

[12 Marks]

[continued ...]

[P. T. O.]

[continue ...]

2.a What is *polymorphism* with respect to object oriented programming?

2.b List down five rules to be satisfied for *method overriding* in Java.

- i) -----

- ii) -----

- iii) -----

- iv) -----

- v) -----

2.c Consider the following class definition to answer the questions below.

```
class GameShape {  
    public void displayShape() {  
        System.out.println("displaying shape");  
    }  
}  
  
class PlayerPiece extends GameShape implements Animatable {  
  
    public void movePiece() {  
        System.out.println("moving game piece");  
    }  
  
    public void animate() {  
        System.out.println("animating...");  
    }  
    // more code  
}
```

[continued ...]

[continue ... 2.c]

Assume the following declarations are made using the above definitions.

```
PlayerPiece player = new PlayerPiece();
Object o = player;
GameShape shape = player;
Animatable mover = player;
```

2.c.i How many objects are in the above declarations?

2.c.ii How many reference variables are in the above declarations?

2.c.iii Which of the above reference variable/s can invoke the displayShape() method?

2.c.iv Explain how Java determines *Type* of objects at runtime.

----- [16 Marks]

3. Consider the following program to answer the questions below.

```
interface Shape {
    int area();
    int perimeter();
}

class Rectangle implements Shape {
    private int width;
    private int height;

    //more codes
}

public class App {
}
```

[continued ...]

[P. T. O.]

[continued ...]

3.a Write down a constructor definition for Rectangle class where it initializes all the instance variables.

3.b Write down getter and setter methods for the instance variable named *width* in the Rectangle class according the *Java Bean* specification.

3.c Write down an implementation of the inherited abstract methods in the Rectangle class.

3.d In order to compile and execute the above program, what could be the suitable file name?

3.e Instantiate an object from Rectangle class where width and height are 10 and 12 respectively.

----- [12 Marks]

4. Consider the following array definition

```
int[ ] numbers = { 12, 4, 5, -1, 19, 16, 5 };
```

[continued ...]

4.a List down five characteristics of an array data structure.

- i) -----

- ii) -----

- iii) -----

- iv) -----

- v) -----

4.b Implement a Java method to remove duplicate numbers from an input array of integers. If duplicate numbers exist, the first occurrence of a number remains while the next occurrences eliminate. For the above array of elements the result is an array of elements { 12, 4, 5, -1, 19, 16}

```
public int[ ] removeDuplicates(int[ ] numbers) {
```

```
-----  
-----  
-----  
-----  
-----  
-----  
-----  
-----  
-----
```

```
}
```

4.c Write a Java method which returns the *smallest even number* of a given array of integers. In the above integer array it is 4.

```
public int findSmallestEvenNumber(int[ ] numbers ) {
```

```
-----  
-----
```

[continue ... 4.c]

}

[20 Marks]

5.a Explain the following statement related to Java bytecode.

"Compiled Java bytecode files are architecture-independent, but Virtual Machine dependent."

5.b Explain the necessity of defining methods in programming.

5.c Write a Java program to count the number of vowels in a given English string having both capital and simple letters (a, e, i, o, u, A, E, I, O, U).

Example: input argument: 'Internationalization', returns 10;

[continued ...]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

[continued ...]

[P. T. O.]

PART B

A mobile service providing company maintains the monthly call records in a text file for each consumer. The tariff is applied according to the tariff plan given below. The cost is calculated minute basis that is cost for 1 minute and 1 second will be considered as 2 minutes.

Tariff Plan:

First two minutes, one minute costs Rs. 3.00

Next 3 minutes, one minute costs Rs 3.50

For the rest, one minute costs Rs 4.00

Name of the input text file is formed using mobile number, year and month. Three fields are separated using underscore (Example: 0799112224_2017_06.txt). The content fields of text files are separated by commas and they are date, call start time, call end time, and duration. Call duration is formatted as hours: minutes: seconds. A sample text file is given below.

```
2017-06-01, 12:31:02, 12:32:32, 0:1:32
2017-06-01, 14:35:12, 14:38:22, 0:3:10
2017-06-02, 07:12:13, 07:12:44, 0:0:31
...
2017-06-30, 17:44:11, 17:51:41, 0:7:30
```

File: 0799112224_2017_06.txt

Write down a complete Java program to calculate the mobile phone bill for a consumer in a given month. The output should be formatted as below (the corresponding information will replace dots).

Mobile Phone Number:

Year:

Month:

Total Call Hours:

Total Cost:

[20 Marks]

--- End ---



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
 END SEMESTER EXAMINATION (SEMESTER II) - 2015/2016
 CS102 - PROGRAMMING TECHNIQUES (3 CREDITS)

Answer **ALL** Questions

Time Allowed: **THREE Hours**

- 1) a) What is the difference between JDK and JRE? [2 Marks]
- b) What are the primary data types supported by Java? [4 Marks]
- c) What is 'autoboxing and unboxing' in Java? [2 Marks]
- d) What are the differences between String, StringBuilder and StringBuffer in Java? [3 Marks]
- e) Why character array is better than String for storing password in Java? [4 Marks]

- 2) a) Describe break and continue flow control structures in Java with suitable examples. [2 Marks]
- b) Write down the list of characteristics of constructors in a Java class. [4 Marks]
- c) Explain *this* and *super* keywords with respect to constructor invocation. [3 Marks]
- d) Can a constructor be defined with static, abstract and final modifiers?. Justify your answer. [6 Marks]

- 3) a) Explain when to write class-methods (with *static* keyword) in programming? [3 Marks]
- b) What is meant by 'method signature'? [2 Marks]
- c) Write a Java program to find the second highest number in an integer array? [5 Marks]
- d) Write a java class-method to remove vowels from an English string?
 Example: input argument: 'Satisfied', return string: 'Stsfd' [5 Marks]
- e) Write a method to print first 20 prime numbers.
 Hint: A prime number is a whole number greater than 1, whose only two whole-number factors are 1 and itself. [5 Marks]

[Continued ...]

[P.T.O.]

- 4) a) Explain briefly similarities and differences between *abstraction* and *encapsulation*.
[3 Marks]
- b) Briefly explain polymorphism with a suitable example.
[3 Marks]
- c) Write down the conditions to be satisfied in method overriding. You may consider access modifiers, special modifiers, return types and throws clauses.
[3 Marks]
- d) Can a constructor be overridden?. Justify your answer.
[2 Marks]
- e) What is Java Exception Handling?. What is the difference between Errors, Unchecked exceptions and Checked exceptions?
[2 Marks]
- f) What is the difference between 'throw' and 'throws' in Java Exception Handling ?
[2 Marks]
- 5) a) Explain briefly the necessity of having enumerations (enums) in programming languages.
[2 Marks]
- b) Explain how do generics reduce runtime exceptions in Java.
[2 Marks]
- c) Explain the difference between List and Set data structures in Java ?
[2 Marks]
- d) When is it necessary to override *equals* and *hashCode* methods in a Java class?
[3 Marks]
- e) Briefly describe the term *streams* in Java.
[3 Marks]
- f) Explain the differences between byte streams and character readers in Java.
[3 Marks]
- 6) a) Write Java code snippets for the following requirements.
i) Append a text to an existing file, named data.txt.
ii) Read a UTF-8 encoded text file, named encoded.txt.
[4 Marks]
- b) The following text file (patient.txt) contains information about patients' blood glucose levels, in which column one represents patient name and next three columns are three readings in different time intervals.

[Continued ...]

[... 6 .b) Continue]

patient.txt

S.M. Rathnayake, 112, 145, 124

P. Edirisingha, 178, 145, 133

Sachith Ruberu, 92, 100, 123

Menaka Wijesingha, 87, 92, 92

Savitri Ekanayake, 178, 155, 167

Read the text file and print the average blood glucose level of each patient against the patient name and if the average blood glucose level is greater than 150, print "At Risk" after the average value.

e.g. Name: P. Edirisingha, Average Blood Glucose Level : 152.0 mg/dl (At Risk)

[16 Marks]



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
 END SEMESTER EXAMINATION (SEMESTER II) - 2014/2015
 CS102 - PROGRAMMING TECHNIQUES (3 CREDITS)

Answer **ALL** Questions

Time Allowed: **TWO Hours**

- 1) a) Why do programming languages have several data types?
[2 Marks]
- b) Briefly describe the differences between *float* and *double* data types in Java.
[4 Marks]
- c) State clearly the complete set of rules used in defining identifiers in a Java program.
[6 Marks]
- d) What are the main differences between program arguments and virtual machine arguments in Java?
[2 Marks]
- e) What is meant by thread-safe data types / structures in Java? Give two examples for thread-safe and non-thread safe data types / structures.
[4 Marks]
- f) Explain graphically *called method* and *calling method* in terms of method calling.
[2 Marks]
- 2) a) Describe all the flow control structures available in Java with suitable examples.
[6 Marks]
- b) Explain the differences between *class variables* and *instance variables* with suitable examples. You must clearly show how these variables types are accessed from outside of the class.
[4 Marks]
- c) Write down the list of characteristics of constructors in a Java class.
[4 Marks]
- d) Explain briefly similarities and differences between *abstraction* and *encapsulation*.
[4 Marks]
- e) Explain why global constants are defined as *public static final* in a class rather in an interface?
[2 Marks]
- f) Write down the conditions to be satisfied in method overriding. You may consider access modifiers, special modifiers, return types and throws clauses.
[5 Marks]

[Continued...]

[P.T.O.]

g) Does the reflection API of Java break data encapsulation?, Justify your answer.

[5 Marks]

3) a) Explain briefly the necessity of having enumerations (enums) in programming languages.

[4 Marks]

b) Explain how generics reduce runtime exceptions in Java.

[4 Marks]

c) What are the main differences between checked and unchecked exceptions? Give two examples for each type.

[4 Marks]

d) Write a suitable example to explain the advantages of using variable arguments (varargs) in Java.

[4 Marks]

e) When is it necessary to override *equals* and *hashCode* methods in a Java class?

[3 Marks]

f) What are the data structures available under Java Collection framework? Also compare List, LinkedList, Vector and ArrayList in the java.util package.

[6 Marks]

4) a) Briefly describe the term *streams* in Java.

[3 Marks]

b) Explain the differences between byte readers and character readers in Java.

[6 Marks]

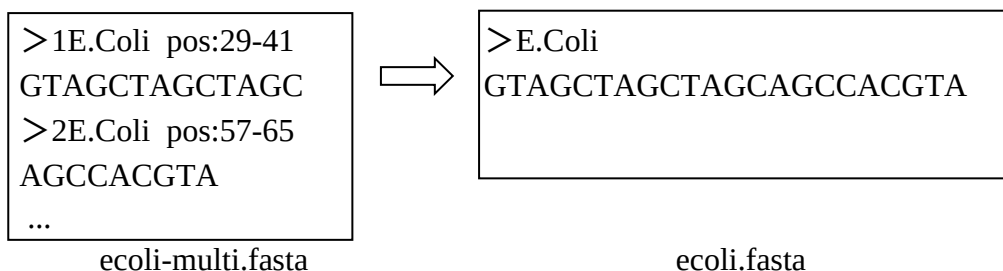
c) Write Java code snippets for the following requirements.

i) Append a text to an existing file, named data.txt.

ii) Read a UTF-8 encoded text file, named encoded.txt.

[6 Marks]

d) It is required to convert multi fasta formatted file (ecoli-multi.fasta) to a fasta formatted file (ecoli.fasta), where all the sequence segments are concatenated to a single sequence. Write a Java program to read ecoli-multi.fasta file and generate the ecoli.fasta file. Note: input file may contain more than two sequence segments.



[10 Marks]



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
MID SEMESTER EXAMINATION (SEMESTER 1) - 2012/2013

Programming Techniques - CS 102

Answer All Questions

Time Allowed: 60 minutes

Q1. Select the most appropriate answer.

[10*2 marks]

1. What is not correct about the JAVA language?
 - a) Java does not support procedural programming
 - b) Java is purely object oriented and supports multiple inheritance
 - c) Java programs can be executed on any operating system
 - d) Java provides a garbage collector

2. The process of hiding internal data from the outside world, and accessing it only through publicly exposed methods is known as
 - a) Abstraction
 - b) Inheritance
 - c) Information hiding
 - d) Encapsulation

3. What is true about abstract classes?
 - a) They can be defined with the abstract key word as well as without the abstract key word
 - b) Abstract class can not be used as the base class of an inheritance hierarchy
 - c) It can not have fully defined methods
 - d) Abstract methods defined in an abstract class must be refined by its subclasses

4. Instance variables of a class
 - a) Can be accessed by static methods
 - b) Can be accessed through the class name
 - c) Belong to objects of the class rather than the class
 - d) Can be declared as private as well as public

5. What is not correct about a method?
 - a) Method is a part of a class behavior
 - b) Method must be fully defined in class definitions
 - c) Method could return nothing or a single value
 - d) Method can be declared within a class as well as an interface

6. Which of the following methods can not be included within a class that defines the void count (int a, String s, boolean b) {}?

- a) void count (int a, int b boolean c) {}
- b) void count (int a, String s) {}
- c) void count (int x, String y, boolean z) {}
- d) void count (boolean b) {}

7. Which of the following assignments are correct according to the JAVA numerical type hierarchy?

byte a; short b; double c;

- a) a = c; b = c;
- b) c = ad; c = bD;
- c) a = (byte) c; b = a;
- d) a = (byte) b; b = c;

8. What is true about classes and objects?

- a) A class can have any number of objects and each object could belongs to many classes
- b) Number of attributes of an object must be same as that of its class, but the number of methods of an object could be different than that of its class
- c) Objects communicate each other by means of method invocation
- d) Object can have its own instances

9. Access modifiers are used to

- a) Identify members uniquely
- b) Define the scope of access
- c) Define classes with different members
- d) None of the above

10. What is incorrect about constructors?

- a) Constructor name is same as the class name
- b) One class could have multiple constructors
- c) If you define a parameterized constructor, the class can not have a default constructor
- d) Constructors are used to initialize attributes.

Q2. Find and correct the errors, if any, in the program *Testing.java* given below. If there is an error(s) in a particular line, state the line number and the error. Then explain how to correct the error(s).

1. #include java.io.*;

2. public class New class

NewClass → **spaces x**

3. {

4. private int no;

5. private string name, address, type;

String

```

6.      public double balance, interest;

7.      public void NewClass()
8.      {
9.          no = 0; name = "";
10.         address = ""; Type = "s";    type
11.         bal = 1;    balance
12.     }

13.     double calculateInterest()
14.     {
15.         interest = balance * 0.4;
16.         return interest;
17.     }

18.     public double getBalance()
19.     {
20.         System.out.println(balance);
21.     }    return    balance;

22.     public NewClass(Int a, String s1, String s2, String string s3, double b)
23.     {
24.         no = a;
25.         name = s1;
26.         address = s2;
27.         type = s3;
28.         balance = b;
29.     }

30.     public void display()
31.     {
32.         System.out.println(no + "\n" + name + "\n" + address);
33.     }
34. }

35. public class Testing
36. {
37.     String
38.     public static void main(String args[]) throws IOException
39.     {
40.         int a=25;
41.         String na, ad, ty;
42.         double c = 234.0;

43.         BufferedReader in = new BufferedReader ( new
44.             InputStreamReader(System.in));

```

Enter

```
43.      System.out.println("Enter name, address and type respectively);
44.  na  Na = in.readLine();
45.  ad  Ad = in.readLine();
46.  ty  Ty = in.readLine();

47.      Newclass m = new NewClass (a, na, ad, ty, c);
48.      m.getBalance();
49.      m.display();
50.      double interest = m.calculateInterest();
51.      System.out.println("Interest earn = " + interest);
52.  }    Catch (IOException e) {
53.      e.getMessage();
}
```

[15 Marks]

Q3. Define a class named Employee with following members.

Attributes: empNo (long int)
Name (String)
Category (Boolean)
Salary (double)
Commission(double)

Methods: setValue() - passes parameters to initialize empNo, name, category and salary.
Calcomm() - computes the commission as follows and assigns to commission attribute.
Commission = salary * 25%, if category is YES
Commission = salary * 10%, otherwise
Display() - displays all the attribute values with appropriate messages.

[15 Marks]